TATA STEEL

Crane GatePass & SafetyCheck
Alert Notification System

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FMM-Infrastructure

PROBLEM STATEMENT

High Error Susceptibility

Manual checking of dates daily is prone to oversight. Busy schedules and cognitive load may lead to missed validity milestones.

Reactive Response

If the expiration is identified after the deadline has passed, there would operational delays and grounding of equipment.

PROBLEM STATEMENT

100%

HUMAN RELIANCE

The Compliance Gap

Without automated intervention, the integrity of Crane Gatepass and Safety Checks rests solely on administrative bandwidth.

This introduces a single point of failure in site safety protocols.

THE SOLUTION

Automated Notification System for Crane Gatepass & Safetycheck Expiration Alerts

Cloud Logic

Program runs daily on cloud triggers, removing local dependence.

Proactive Monitoring

Advance warnings enable proactive renewal of Safety Checks and Gatepass.

Centralized Data

Dynamic link between spreadsheets and notification payloads.

INTEGRATION ARCHITECTURE

System Workflow

By linking **Microsoft Excel** with **Power Automate** (or Google Sheets with AppScript), we created a self-sustaining loop.

The system evaluates dates daily and triggers personalized emails based on a defined recipient list within the same excel sheet.

DATA FRAMEWORK

VEHICLE NAME	REGISTRATION NUMBER	GATE PASS VALIDITY	SAFETY CHECK VALIDITY	NOTIFICATION STATUS
450T Crane	MH46CR-1192	04/28/2026	03/13/2027	Monitoring
350T-1	OD34Y-9927	04/21/2026	03/10/2027	Alert Triggered
Trailer Unit	JH05EH-4547	06/20/2026	NA	Monitoring
80T-1	GJ15SV-2323	04/29/2026	09/19/2027	Pending Warning

NOTIFICATION LOGIC

Stage 1: Warning

Notifications are triggered **7 days prior** to expiration. This provides a lead-time buffer for physical inspections and documentation prep.

Stage 2: Critical

Immediate escalations are sent if dates have already expired, ensuring assets are not utilized in a non-compliant state.

IMPACT

Zero Human Error

Cloud triggers ensure no expiration date is ever overlooked.

Proactive Maintenance

7-day windows enable teams to schedule safety checks without disrupting operations.

Resource Optimization

Administrative staff are freed from manual auditing, reallocating hours to high-value tasks.

Audit Readiness

Centralized, updated logs provide instant verification for site safety audits.

THANKYOU